

Aircraft Landing Gear Brake Systems: Standards for Actuation, Thermal Management, and Airworthiness Compliance

Aviation Standards Division

May 24, 2026

Document ID: MRO-LGB-HB-006

Contents

1	Introduction to Aircraft Braking Systems and ATA 32-40 Architecture	2
2	Brake Actuator and Stator-Rotor Assembly Mechanics	4
3	Hydraulic Brake Actuation and Pressure Modulation	6
4	Anti-Skid and Autobrake Control Systems	8
5	Thermal Management and Overheat Protection	10
6	Wheel and Axle Interface: Bearings, Seals, and Torque Specifications	12
7	Wear Indicator Measurement and Brake Life Limits	14
8	Non-Destructive Testing (NDT) of Brake Structural Components	16
9	Brake System Rigging, Bleeding, and Functional Testing	18
10	Parking Brake and Emergency/Pneumatic Brake Architectures	20
11	Taxi, Takeoff, and Rejected Takeoff (RTO) Performance Metrics	22
12	Maintenance Documentation, Logbook Entries, and Airworthiness Directives	24
	Appendix D: Daily Landing Gear & Brake Maintenance Checklist	26

Chapter 1

Introduction to Aircraft Braking Systems and ATA 32-40 Architecture

The fundamental physics of aircraft deceleration during ground roll is governed by the conversion of kinetic energy into thermal energy through friction. The total kinetic energy E_k that the braking system must absorb is defined by the equation $E_k = \frac{1}{2}mv^2$, where m is the maximum certified landing weight of the aircraft and v is the touchdown velocity. For modern transport category aircraft, this energy can exceed 150 Megajoules, necessitating a highly robust and thermally resilient braking architecture. The regulatory framework governing these systems is primarily dictated by 14 CFR Part 25, specifically Section 25.735, which mandates that the braking system must be capable of absorbing the kinetic energy of a rejected takeoff (RTO) at maximum takeoff weight and V_1 speed without structural failure or uncontained fire. Furthermore, the system must demonstrate a specific stopping distance on a dry runway, requiring a minimum average deceleration rate of 0.3g to 0.4g, depending on the aircraft's weight and aerodynamic drag characteristics. The Air Transport Association (ATA) categorizes these systems under ATA Chapter 32, with the specific sub-system for brakes designated as ATA 32-40, encompassing the mechanical actuators, hydraulic metering, anti-skid control, and wear monitoring assemblies.

The architectural layout of the ATA 32-40 system integrates mechanical linkages, hydraulic power distribution, and electronic control units to modulate the clamping force applied to the brake stacks. In a standard multi-wheel main landing gear configuration, each wheel is equipped with a multi-disc brake assembly mounted inboard of the wheel hub. The braking torque is transmitted from the rotating brake discs to the stationary axle via a torque tube, which must withstand immense shear stresses during maximum braking events. The hydraulic pressure required to actuate the brake pistons is routed through flexible hoses that accommodate the vertical stroke and steering articulation of the landing gear strut. The system is designed with complete redundancy; transport category aircraft utilize dual or triple independent hydraulic systems (e.g., Green, Yellow, and Blue) to ensure that a single hydraulic failure does not result in a total loss of braking capability. The isolation of these systems is maintained via shuttle valves and priority valving, which automatically route fluid from the alternate source if the primary system pressure drops below a calibrated threshold, typically 1500 psi in a 3000 psi nominal system.

Deceleration performance is heavily influenced by the runway surface condition, which is

quantified using the Runway Condition Assessment Matrix (RCAM). The effective coefficient of friction μ between the aircraft tire and the runway surface dictates the maximum braking force $F_b = \mu N$, where N is the normal force exerted by the aircraft weight on the braked wheels. On a dry concrete runway, μ can reach values up to 0.8, allowing for aggressive deceleration. However, on wet, snow-covered, or icy runways, μ degrades significantly, dropping to 0.2 or lower, which increases the required stopping distance and necessitates the use of maximum reverse thrust and aerodynamic spoilers to compensate for the reduced brake efficacy. The anti-skid system continuously monitors the wheel slip ratio to maintain the tire operation at the peak of the μ -slip curve, preventing locked-wheel skids that would drastically reduce the friction coefficient and cause severe tire tread degradation. The integration of these mechanical, hydraulic, and electronic subsystems requires precise rigging and continuous airworthiness monitoring to ensure compliance with the stringent safety margins mandated by aviation regulatory authorities.

Chapter 2

Brake Actuator and Stator-Rotor Assembly Mechanics

The core energy-absorbing component of the aircraft brake assembly is the stator-rotor stack, which consists of alternating stationary stators and rotating rotors. Modern commercial aircraft predominantly utilize carbon-carbon composite brake discs due to their superior specific heat capacity, lower mass, and higher energy absorption limits compared to legacy steel brakes. The carbon-carbon matrix is manufactured through a complex process involving the layup of carbon fiber preforms followed by densification via Chemical Vapor Deposition (CVD) or liquid pitch impregnation. This process yields a final material density of approximately 1.75 to 1.85 g/cm³. The friction coefficient of carbon-carbon composites is highly temperature-dependent; it remains relatively low at ambient temperatures but increases significantly as the interface temperature rises to the optimal operating range of 800°C to 1200°C. At these elevated temperatures, the friction coefficient stabilizes between 0.35 and 0.45, providing consistent and predictable braking torque. However, if the temperature exceeds 1500°C, the carbon matrix begins to undergo rapid oxidation, leading to accelerated wear and structural degradation.

In contrast, steel brakes, which are still utilized on many regional and military aircraft, employ steel stator discs mated with rotors lined with sintered iron or copper-based friction materials. Steel brakes operate at significantly lower peak temperatures, typically limited to 800°C to 1000°C, and possess a higher mass, which negatively impacts the aircraft's overall fuel efficiency. The friction coefficient of sintered linings is generally more stable across a wider temperature range but suffers from higher wear rates under severe high-energy stop conditions. The mechanical interface between the rotating rotors and the wheel hub is achieved via drive lugs or splines, which must transmit the entire braking torque without yielding. The shear stress τ on these drive lugs is calculated using the formula $\tau = \frac{T}{A_s \cdot r_m \cdot n}$, where T is the maximum braking torque, A_s is the shear area of a single lug, r_m is the mean radius of the lug engagement, and n is the total number of drive lugs. To prevent galling and excessive wear at the spline interface, the drive lugs are often coated with a dry film lubricant, such as molybdenum disulfide (MoS₂), and must be inspected for step-wear and deformation during routine maintenance checks.

The stationary stators are keyed to the torque tube, which is rigidly bolted to the landing gear axle flange. The torque tube must be engineered to withstand the maximum stall torque of the brake assembly without exceeding the yield strength of the material, typically a high-strength low-alloy (HSLA) steel or a titanium alloy. The torsional rigidity of the torque tube is critical;

excessive angular deflection under load can cause misalignment of the stator stack, resulting in uneven pressure distribution and localized hot spots on the friction surfaces. The polar moment of inertia J of the torque tube's cross-section dictates its resistance to torsional deformation, calculated for a hollow cylindrical tube as $J = \frac{\pi}{2}(r_o^4 - r_i^4)$, where r_o and r_i are the outer and inner radii, respectively. The assembly of the stator-rotor stack requires strict adherence to axial clearance limits to ensure that the discs can expand thermally without binding, while maintaining sufficient overlap to prevent the friction surfaces from disengaging during high-speed rotation. The wear mechanisms of the stack include both mechanical abrasion and thermal oxidation, necessitating precise measurement protocols to determine the remaining service life of the brake assembly.

Chapter 3

Hydraulic Brake Actuation and Pressure Modulation

The actuation of the brake stacks is achieved through hydraulic pressure applied to a series of pistons housed within the brake actuator cylinder. Transport category aircraft typically utilize high-pressure hydraulic systems operating at a nominal pressure of 3000 psi (20.7 MPa), with newer airframes transitioning to 5000 psi (34.5 MPa) architectures to reduce the physical size and mass of the actuators and plumbing. The brake metering valve, located in the landing gear bay or integrated into the rudder pedal assembly, modulates the system pressure based on pilot input or autobrake commands. The flow dynamics through the metering valve orifices are governed by the standard orifice equation $Q = C_d A \sqrt{\frac{2\Delta P}{\rho}}$, where Q is the volumetric flow rate, C_d is the discharge coefficient, A is the variable orifice area, ΔP is the pressure differential, and ρ is the hydraulic fluid density. The precise modulation of this orifice area is critical to providing linear pedal feel and preventing abrupt pressure spikes that could induce structural shock loads on the landing gear trunnion.

The hydraulic fluid utilized in these systems, typically a fire-resistant phosphate ester conforming to SAE AS1241 (e.g., Skydrol or HyJet), exhibits a high bulk modulus β , which is a measure of the fluid's resistance to compression. The bulk modulus is defined as $\beta = -V \frac{dP}{dV}$, where V is the fluid volume and dP/dV is the rate of pressure change with respect to volume. A high bulk modulus ensures rapid pressure transmission and minimal pedal sponginess; however, the entrainment of even minute quantities of air drastically reduces the effective bulk modulus, leading to delayed brake response and excessive pedal travel. To mitigate this, the brake actuators are equipped with automatic bleed valves that purge trapped air during the initial pressurization cycles. The actuator pistons are sealed using specialized elastomeric seals, typically Ethylene-Propylene Diene Monomer (EPDM) for phosphate ester fluids, which must maintain a leak-tight seal under extreme pressures and temperatures. The allowable external leakage rate for a brake actuator is strictly limited by the manufacturer, generally not exceeding one drop per minute at maximum system pressure, while internal leakage across the piston seals must be virtually zero to maintain clamping force during prolonged parking brake applications.

The physical layout of the brake actuator cylinders varies between segmented and annular designs. Segmented actuators utilize multiple discrete pistons arranged circumferentially around the pressure plate, requiring a complex internal manifold to ensure equal pressure distribution to each piston. Annular actuators, conversely, utilize a single, continuous ring-shaped piston that

applies uniform pressure across the entire brake stack, simplifying the hydraulic routing and reducing the risk of uneven clamping. The response time of the hydraulic actuation system is a critical safety parameter; the time elapsed from the initial pedal displacement to the achievement of 90% of the commanded brake pressure must not exceed 0.2 seconds. This rapid response is essential for the anti-skid system to effectively modulate brake torque and prevent wheel lock-up during low-friction runway conditions. The hydraulic lines routing fluid to the brake actuators must be constructed from high-strength stainless steel or titanium, utilizing flexible hoses only where necessary to accommodate gear retraction and steering articulation, with strict minimum bend radii enforced to prevent hose collapse and flow restriction.

Chapter 4

Anti-Skid and Autobrake Control Systems

The anti-skid system is a closed-loop electro-hydraulic control mechanism designed to maximize braking efficiency while preventing wheel lock-up and tire blowouts. The system relies on wheel speed transducers, typically variable reluctance or active Hall-effect sensors, mounted on the wheel axle to measure the rotational velocity of each braked wheel. The anti-skid control unit (ABCU) continuously calculates the wheel slip ratio λ , defined as $\lambda = \frac{v_a - \omega r}{v_a}$, where v_a is the aircraft's true ground speed, ω is the wheel angular velocity, and r is the loaded tire radius. The coefficient of friction between the tire and the runway peaks at a slip ratio of approximately 10% to 15%; if the slip ratio exceeds this threshold, the wheel enters an unstable skid condition, and the friction coefficient drops precipitously. The ABCU utilizes adaptive proportional-integral-derivative (PID) control algorithms operating at frequencies exceeding 100 Hz to modulate the anti-skid valves, rapidly releasing and reapplying brake pressure to maintain the slip ratio at the optimal peak.

The autobrake system provides automated, consistent deceleration rates during landing and rejected takeoff scenarios, reducing pilot workload and ensuring optimal utilization of the runway length. The autobrake selector allows the pilot to choose predefined deceleration profiles, typically categorized as Low, Medium, and Maximum. The Low setting is optimized for dry runways and passenger comfort, targeting a deceleration rate of approximately 0.15g to 0.20g. The Medium setting provides a more aggressive deceleration of 0.25g to 0.30g, suitable for wet runways or shorter landing distances. The Maximum setting commands the highest possible deceleration, limited only by the anti-skid system's ability to maintain tire traction, and is primarily utilized during rejected takeoffs. The autobrake system interfaces directly with the aircraft's air/ground sensing logic, inertial reference system, and thrust reverser deployment sensors to ensure that autobrake activation occurs only when the main landing gear is firmly on the ground and the thrust levers are at idle.

The electrical wiring and signal transmission for the anti-skid and autobrake systems must adhere to stringent electromagnetic interference (EMI) and lightning protection standards. The wheel speed sensor signals are highly susceptible to noise induced by the high-current power cables routed along the landing gear strut. To mitigate this, the sensor wiring utilizes shielded twisted pair (STP) cables, with the shield grounded at a single point to prevent ground loops. The characteristic impedance of the transmission lines and the input impedance of the ABCU

must be matched to prevent signal reflection and attenuation over the long cable runs from the wheels to the equipment bay. Furthermore, the wiring harnesses within the wheel well are subjected to extreme environmental conditions, including high-velocity debris impact, hydraulic fluid exposure, and thermal cycling. Consequently, the cables are encased in abrasion-resistant braided sleeving and routed with strict minimum clearances from the brake heat shields and hydraulic actuators to prevent chafing and subsequent short circuits, which could result in an uncommanded brake application or a total loss of anti-skid protection.

Chapter 5

Thermal Management and Overheat Protection

The absorption of massive kinetic energy during braking results in extreme thermal loading on the brake stack, with peak temperatures frequently exceeding 1500°C during a maximum energy rejected takeoff. This intense heat generation necessitates robust thermal management strategies to protect the surrounding landing gear components, hydraulic lines, and wheel assemblies from catastrophic thermal degradation. The primary mechanism for heat dissipation from the carbon-carbon or steel brake stack is thermal radiation, governed by the Stefan-Boltzmann law $q = \epsilon \sigma AT^4$, where q is the radiated heat flux, ϵ is the emissivity of the brake surface, σ is the Stefan-Boltzmann constant, A is the surface area, and T is the absolute temperature. To maximize radiative cooling, the external surfaces of the brake housings and torque tubes are often treated with high-emissivity thermal barrier coatings (TBCs). Additionally, convective cooling occurs as ambient air flows through the wheel well and the open rotor-stator stack during taxi and takeoff roll, though this effect is minimal during the initial high-heat soak period immediately following a heavy stop.

To protect the wheel and tire assembly from the intense radiant heat emitted by the brake stack, heavy-duty thermal shields are installed between the brake and the inboard wheel flange. These shields are typically constructed from multi-layered Inconel or stainless steel foil separated by insulating ceramic fiber mats, providing a high degree of thermal resistance. Despite these shields, the wheel rim temperature can still rise significantly during a severe braking event. To prevent a catastrophic tire explosion caused by the overpressurization of the nitrogen tire inflation gas due to thermal expansion, the wheels are equipped with thermal fuse plugs. These plugs contain a eutectic alloy core designed to melt at a specific calibrated temperature, typically between 160°C and 180°C (320°F to 356°F). When the alloy melts, the plug core is ejected by the tire pressure, safely deflating the tire in a controlled manner and eliminating the risk of a high-energy shrapnel event. Maintenance personnel must strictly verify the correct part number and temperature rating of the thermal fuses during wheel installations, as the installation of an incorrect fuse could result in premature deflation or failure to relieve pressure during an overheat condition.

The cooling time of the brake assembly is a critical operational parameter, particularly for aircraft conducting short-turnaround flights where the brakes may not have sufficient time to dissipate the heat from the previous landing before the next takeoff. If the brakes are excessively

hot at the start of the takeoff roll, their capacity to absorb the kinetic energy of a potential rejected takeoff is severely compromised, increasing the risk of brake failure and uncontained fire. To monitor this, modern aircraft are equipped with Brake Temperature Monitoring Systems (BTMS), which utilize thermocouples embedded within the stationary stators to provide real-time temperature data to the flight deck. The flight crew uses this data to enforce mandatory cooling delays and to calculate derated takeoff speeds if necessary. Furthermore, the wheel wells of transport category aircraft are equipped with fire detection and extinguishing systems, utilizing ultraviolet/infrared (UV/IR) optical flame detectors and Halon 1301 or environmentally acceptable clean agent extinguishers, to suppress any hydraulic fluid or grease fires that may ignite due to contact with the superheated brake components.

Chapter 6

Wheel and Axle Interface: Bearings, Seals, and Torque Specifications

The wheel assembly rotates on the landing gear axle via a set of high-precision, heavy-duty roller bearings, typically tapered roller bearings or cylindrical roller bearings, designed to support the immense radial loads of the aircraft weight and the axial thrust loads generated during cornering and crosswind landings. The inner and outer bearing cones are pressed into the wheel hub with a strict interference fit to prevent the bearing cups from spinning (creeping) within the hub bore during operation, which would cause severe galling and structural weakening of the aluminum or magnesium wheel forging. The lubrication of these bearings is critical; they are packed with a specialized aviation-grade grease, such as MIL-PRF-23827, which must maintain its viscosity and anti-wear properties across a temperature range from -54°C to 150°C . The grease must be applied in a precise quantity, typically filling 30% to 50% of the bearing's internal void volume, to prevent churning and subsequent overheating during high-speed rotation.

The adjustment of the wheel bearing preload, or axial endplay, is a highly critical maintenance procedure that directly impacts the bearing's fatigue life and the structural integrity of the wheel assembly. Excessive endplay allows the wheel to wobble, inducing cyclic impact loads on the bearing rollers and the axle spindle, while excessive preload generates immense internal friction, leading to rapid grease degradation, overheating, and catastrophic bearing seizure. The acceptable axial endplay for main landing gear wheels is strictly defined by the OEM, typically ranging from 0.004 to 0.010 inches (0.10 to 0.25 mm). This endplay is achieved by tightening the inner axle nut to a specified seating torque, rotating the wheel to seat the bearing rollers, and then backing off the nut by a precise number of castellations before securing it with a new cotter pin. The outer retaining nut is then torqued to a much higher value, often exceeding 400 to 600 ft-lbs, to clamp the entire bearing assembly securely against the axle shoulder.

Sealing the wheel hub to retain the bearing grease and exclude moisture, runway debris, and hydraulic fluid is achieved through a combination of elastomeric lip seals and labyrinth seals. The primary grease retention seal is typically a spring-loaded nitrile or fluorocarbon lip seal that rides on a hardened steel wear sleeve pressed onto the axle. The condition of this wear sleeve must be meticulously inspected during every tire change; any scoring, pitting, or corrosion on the sleeve surface will rapidly destroy the seal lip, leading to grease starvation and bearing failure. Additionally, the inboard side of the wheel hub incorporates an O-ring seal that interfaces with the axle to prevent the ingress of water and de-icing fluids into the bearing cavity. The wheel

halves, which are bolted together to enclose the tire, are sealed using a specialized O-ring or a continuous bead of RTV silicone sealant to maintain the integrity of the tubeless tire air seal. The tie-bolts securing the wheel halves must be torqued in a strict star-pattern sequence to the manufacturer's specified value, typically 150 to 250 ft-lbs, and secured with safety wire to prevent loosening under the severe vibrational environment of the landing gear.

Chapter 7

Wear Indicator Measurement and Brake Life Limits

The continuous monitoring of brake wear is essential to ensure that the stator-rotor stack retains sufficient thermal mass to absorb the kinetic energy of a maximum-energy rejected takeoff without exceeding the structural or thermal limits of the materials. For carbon-carbon brakes, wear is measured using mechanical wear indicator pins that are threaded into the pressure plate and extend through the backplate of the brake housing. As the friction surfaces of the carbon discs wear away, the pressure plate moves further inboard during brake application, pushing the wear indicator pin outward. The exposed length of the pin is measured using a calibrated go/no-go gauge or a precision ruler while the brakes are fully applied. It is imperative that the measurement is taken under applied hydraulic pressure; measuring the pin with the brakes released will yield an erroneously low reading due to the axial clearance (drag clearance) between the un-actuated rotors and stators. The minimum allowable pin protrusion is strictly defined in the Aircraft Maintenance Manual (AMM), and if the pin extends beyond this limit, the entire brake assembly must be removed and replaced before the next flight.

In addition to mechanical wear pins, advanced brake monitoring systems utilize ultrasonic thickness measurement or linear variable differential transformers (LVDTs) integrated into the brake actuator to provide continuous, real-time wear data to the aircraft's central maintenance computer. For steel brakes, wear is typically measured by checking the total thickness of the brake stack using a specialized caliper or depth gauge inserted through inspection ports in the torque tube. The wear limits for steel brakes are defined by the minimum allowable thickness of the friction linings; if the lining wears down to the backing plate, the steel-on-steel contact will result in a catastrophic loss of friction and severe damage to the mating discs. Furthermore, the heat sink materials are subject to degradation mechanisms other than mechanical wear. Carbon brakes suffer from thermal oxidation, where the carbon matrix reacts with atmospheric oxygen at high temperatures, converting the solid carbon into carbon dioxide gas. This oxidation can significantly reduce the mass and thermal capacity of the brake even if the mechanical wear pins indicate acceptable thickness.

The replacement and overhaul of brake assemblies are governed by strict life limits and mixing restrictions. Carbon brake discs from different manufacturing batches or with different CVD densification profiles must never be mixed within the same brake stack, as discrepancies in the friction coefficients and thermal expansion rates will cause uneven pressure distribution, severe

vibration (brake squeal), and accelerated localized wear. When installing a new or overhauled brake assembly, the technician must verify the part number, serial number, and the accumulated life limits (measured in landings or kinetic energy units) against the component's historical log card. The installation procedure requires the careful alignment of the drive lugs and torque tube splines, often necessitating the rotation of the wheel to mesh the components without applying excessive force that could chip the brittle carbon edges. Following installation, a functional check of the brake actuation and anti-skid system must be performed, and the wear indicator readings must be documented in the aircraft's permanent maintenance records to establish the baseline for future wear trend monitoring.

Chapter 8

Non-Destructive Testing (NDT) of Brake Structural Components

The structural integrity of the load-bearing components within the landing gear brake and wheel assembly is continuously subjected to high-cycle fatigue, impact loading, and severe thermal gradients. To detect sub-surface flaws and prevent catastrophic in-flight failures, rigorous Non-Destructive Testing (NDT) protocols are mandated during heavy maintenance checks and component overhauls. The torque tube, which transmits the braking torque from the stators to the axle, is highly susceptible to fatigue cracking at the splined interfaces and the mounting flange bolt holes. Eddy Current Testing (ECT) is the primary method utilized to inspect these critical areas. The ECT probe induces an alternating magnetic field into the conductive steel or titanium material, and any discontinuities, such as micro-cracks or corrosion pitting, disrupt the resulting eddy currents, altering the impedance of the probe coil. The inspection must be performed using calibrated reference standards with known artificial flaws to ensure the equipment's sensitivity is sufficient to detect cracks as small as 0.050 inches in length.

For ferromagnetic components, such as steel axles, drive lugs, and bearing cones, Magnetic Particle Inspection (MPI) is the standard NDT technique. The component is magnetized using a direct current yoke or a central conductor, and a fine suspension of fluorescent magnetic particles is applied to the surface. If a surface or near-surface crack is present, the magnetic flux lines will leak out of the material, attracting the particles and forming a highly visible indication under ultraviolet (black) light. The magnetization must be applied in at least two orthogonal directions to ensure that cracks of any orientation are detected. Prior to MPI, the components must be meticulously cleaned and demagnetized to prevent the accumulation of abrasive metallic debris during subsequent operation. For the aluminum or magnesium wheel hub forgings, Fluorescent Penetrant Inspection (FPI) is utilized to detect surface-breaking fatigue cracks, particularly in the high-stress areas surrounding the tie-bolt holes and the bead seat flanges. The FPI process involves applying a highly penetrating fluorescent dye, allowing a specified dwell time, removing the excess penetrant, and applying a developer to draw the trapped dye out of any flaws.

Ultrasonic Testing (UT) is employed to inspect the internal structure of thick cross-sections, such as the wheel hub flange and the axle spindle, for internal voids, inclusions, or delaminations. High-frequency sound waves are introduced into the material via a piezoelectric transducer, and the reflections from internal boundaries and flaws are analyzed on an oscilloscope display. The sound velocity and attenuation characteristics of the specific alloy must be precisely calibrated

to ensure accurate depth sizing of any detected indications. For carbon-carbon brake discs, UT is utilized to assess the internal density and detect any delamination between the CVD layers or severe internal oxidation that may not be visible on the external surfaces. The acceptance criteria for all NDT inspections are strictly defined by the OEM engineering drawings and regulatory advisory circulars; any component exhibiting a flaw that exceeds the allowable limits must be immediately scrapped and destroyed to prevent its inadvertent return to service, and the NDT results must be permanently recorded and signed off by a certified Level II or Level III NDT inspector.

Chapter 9

Brake System Rigging, Bleeding, and Functional Testing

The precise mechanical rigging of the brake pedal linkages and hydraulic metering valves is fundamental to ensuring proper brake response, adequate pedal travel, and the prevention of unintended brake drag. The rudder pedal assembly must be rigged to the neutral position with zero pre-load on the brake master cylinders. If the linkage is rigged too tightly, the master cylinder pistons will not fully return to their rest position, blocking the return port and trapping high-pressure fluid in the brake actuators. This condition, known as brake drag, causes continuous friction, rapid wear, and severe overheating of the brake stack during taxi and flight. The rigging procedure involves measuring the free-play at the top of the rudder pedals, which must typically fall within a strict tolerance of 0.5 to 1.0 inches, and adjusting the turnbuckles on the push-pull rods until the specified clearances are achieved. The physical travel of the pedals from the neutral position to the mechanical stop must also be verified to ensure that the master cylinders can achieve full stroke and deliver maximum system pressure.

Following any maintenance that opens the hydraulic system, such as the replacement of a brake actuator or a hydraulic line, the system must be thoroughly bled to remove all entrained air. Air in the hydraulic system is highly compressible, leading to a spongy pedal feel, delayed brake response, and a severe reduction in the anti-skid system's ability to modulate pressure rapidly. Bleeding is typically performed using a pressure bleeding cart that supplies clean, filtered hydraulic fluid to the system reservoir at a pressure of 50 to 100 psi. The technician opens the bleed valves located at the highest points of the brake actuators and the hydraulic lines, allowing the fluid to flow until a solid, bubble-free stream is observed. The sequence of bleeding must follow the manufacturer's specified routing path, usually starting from the components furthest from the reservoir and working inboard. In some modern aircraft, the flight control computers execute an automated bleed sequence, rapidly cycling the anti-skid valves and autobrake servos to purge trapped air from the complex internal manifolds.

Functional testing of the brake system is the final verification step before returning the aircraft to service. This testing includes both static ground tests and, if necessary, dynamic taxi tests. During the static test, the hydraulic system is pressurized, and the brakes are applied to verify that the pressure builds up to the commanded level within the specified response time (e.g., less than 0.2 seconds) and that the pressure holds without decaying beyond the allowable leakage limits over a specified dwell time. The anti-skid system is tested by jacking the aircraft,

spinning the wheels using a drive motor or the aircraft's engines, and simulating a locked-wheel condition to verify that the anti-skid valves correctly dump the brake pressure. The autobrake system is verified by commanding the various deceleration settings and confirming that the correct hydraulic pressure profiles are output to the actuators. All functional test parameters, including pressure hold times, response times, and anti-skid cycle counts, must be meticulously recorded in the maintenance logbook, and any discrepancies must be resolved and re-tested before the aircraft is approved for flight.

Chapter 10

Parking Brake and Emergency/Pneumatic Brake Architectures

The parking brake system is designed to securely hold the aircraft stationary during ground operations, loading, and extended parking periods. In most transport category aircraft, the parking brake is actuated by pulling a mechanical handle in the cockpit, which mechanically locks the brake metering valves in the applied position or isolates a dedicated volume of high-pressure hydraulic fluid in the brake lines. The holding force of the parking brake is dependent on the system pressure at the time of application and the integrity of the check valves and actuator seals. To prevent the aircraft from rolling due to slow internal hydraulic leakage over time, the parking brake system is often integrated with a hydraulic accumulator that provides a reserve volume of pressurized fluid to compensate for minor leaks and maintain the clamping force. The accumulator precharge pressure must be verified regularly, as a degraded nitrogen charge will result in a rapid loss of parking brake holding capability. Furthermore, when parking on steep slopes or in high-wind conditions, wheel chocks are mandatory, as the parking brake alone may not provide sufficient holding torque to overcome the gravitational or aerodynamic forces acting on the aircraft.

In the event of a total failure of the primary hydraulic systems, aircraft are equipped with emergency braking architectures to ensure that the aircraft can be safely stopped. These emergency systems typically utilize a dedicated, isolated hydraulic accumulator or an air-over-hydraulic pneumatic backup system. The pneumatic backup system draws high-pressure air from the aircraft's pneumatic manifold or a dedicated emergency air bottle and routes it to an air-over-hydraulic booster. This booster utilizes the pneumatic pressure to drive a hydraulic piston, generating the necessary fluid pressure to actuate the main brake assemblies. The capacity of the emergency accumulator or air bottle is strictly calculated to provide a minimum number of full brake applications, typically six to eight, from the maximum precharge pressure down to the minimum pressure required to generate adequate braking torque. The gas law equation $P_1 V_1^\gamma = P_2 V_2^\gamma$ is utilized to size the accumulator, where P is pressure, V is volume, and γ is the specific heat ratio of the nitrogen gas.

The routing and valving of the emergency brake system incorporate shuttle valves that automatically prioritize the emergency fluid source when the primary system pressure drops below a

critical threshold. These shuttle valves must be inspected and functionally tested during heavy maintenance checks to ensure that they shift freely and do not allow cross-bleeding between the primary and emergency systems. The cockpit indications for the emergency brake system, including accumulator pressure gauges and low-pressure warning lights, must be fully operational. Pilots are trained to use the emergency brakes judiciously, as the limited number of applications and the lack of anti-skid protection in some emergency architectures require precise pedal modulation to prevent wheel lock-up and tire blowouts. The maintenance documentation must explicitly record the leak-down test results of the emergency accumulators, verifying that the pressure decay rate does not exceed the maximum allowable limits specified in the AMM, ensuring that the emergency braking capacity is fully available when required.

Chapter 11

Taxi, Takeoff, and Rejected Takeoff (RTO) Performance Metrics

The operational performance of the aircraft braking system is most severely tested during the taxi phase and the potential for a high-speed Rejected Takeoff (RTO). During taxi, the frequent application of brakes to control ground speed and negotiate turns results in significant cumulative wear, particularly for carbon brakes, which exhibit higher wear rates at lower temperatures (below 400°C) due to the mechanical abrasion of the un-sintered surface layers. To mitigate taxi wear, flight crews are trained to utilize "intermittent braking" rather than "drag braking." Intermittent braking involves applying the brakes firmly to decelerate the aircraft to a lower speed threshold, releasing them to allow the aircraft to coast, and reapplying them only when the upper speed threshold is reached. This technique allows the brake temperatures to rise into the optimal range where the carbon matrix is more resistant to mechanical wear, significantly extending the service life of the brake stack.

The most extreme energy absorption event the brake system is designed to withstand is the Rejected Takeoff (RTO) at the maximum certified takeoff weight and at the V_1 decision speed. During an RTO, the engines are spooled down to idle, the thrust reversers are deployed, and the brakes must absorb the vast majority of the aircraft's kinetic energy, which can exceed 100 Megajoules per wheel on large wide-body aircraft. The kinetic energy calculation for the RTO is defined as $E_{RTO} = \frac{1}{2}m_{MTOW}V_1^2$, where m_{MTOW} is the maximum takeoff weight and V_1 is the critical engine failure recognition speed. The V_1 speed is often limited not by the runway length or aerodynamic performance, but by the maximum certified brake energy limit (E_{max}). If the calculated kinetic energy exceeds the brake's E_{max} , the V_1 speed must be reduced, which consequently reduces the maximum allowable takeoff weight or requires a longer runway to achieve the lower rotation speed.

Following an RTO, the brake temperatures will soar to extreme levels, often exceeding 1500°C to 2000°C, causing the carbon discs to glow bright orange and generating massive amounts of thermal radiation. The structural integrity of the brake components, the wheel thermal fuses, and the tire inflation gas are all pushed to their absolute design limits. The flight crew must not approach the landing gear until the brakes have sufficiently cooled and the thermal fuses have been visually verified, as a delayed tire explosion or brake structural collapse can occur several minutes after the aircraft has come to a halt. The Brake Temperature Monitoring System (BTMS) provides critical data to the crew, indicating when it is safe to taxi the aircraft to

the gate or when emergency services must be dispatched to monitor the gear for fire. Post-RTO maintenance actions are severe; the entire brake and wheel assembly must be removed, completely disassembled, and subjected to rigorous NDT and dimensional inspections to assess the extent of thermal degradation, oxidation, and structural yielding before any components can be returned to service.

Chapter 12

Maintenance Documentation, Logbook Entries, and Airworthiness Directives

The legal and regulatory framework governing the maintenance, inspection, and modification of aircraft braking systems is strictly enforced under 14 CFR Part 43. Every maintenance action, from a routine tire change and brake wear measurement to the complete replacement of a brake actuator, must be meticulously documented in the aircraft's permanent maintenance logbook. The logbook entry must provide a comprehensive description of the work performed, explicitly referencing the specific maintenance manual chapters, engineering drawings, and service bulletins utilized. The entry must also record the exact torque values applied to critical fasteners, the part numbers and serial numbers of the installed components, and the results of any functional tests or measurements, such as the brake wear pin readings and the wheel bearing endplay. The certifying mechanic or inspector must sign the entry, print their name, and provide their FAA certificate number or the certificate number of the authorized repair station, thereby legally attesting that the work was performed in accordance with approved data and that the aircraft is in an airworthy condition.

Airworthiness Directives (ADs) issued by the FAA or EASA are mandatory regulatory actions that dictate specific inspections, modifications, or life limits to correct known unsafe conditions in the braking system. Compliance with ADs is non-negotiable and must be tracked rigorously by the operator's maintenance control department. For example, an AD may mandate the recurring eddy current inspection of the brake torque tube splines every 2000 flight cycles to detect fatigue cracking, or the immediate replacement of a specific batch of brake actuators due to a manufacturing defect in the internal piston seals. The logbook entry for an AD must explicitly state the AD number, the specific paragraph complied with, the method of compliance, and the date and cycle count at which the compliance was achieved. If the AD is a recurring inspection, the entry must also state the time or cycle count at which the next inspection is due. Failure to comply with an AD renders the aircraft legally unairworthy, and operating the aircraft in this condition is a severe violation of federal aviation regulations.

The tracking of Life-Limited Parts (LLPs) within the brake and wheel assembly is another critical aspect of airworthiness documentation. Components such as the wheel hub forging, the axle spindle, and specific high-stress structural fittings are assigned a strict fatigue life limit, measured in flight cycles or landings. Once a component reaches its certified life limit, it must be permanently removed from service and mutilated to prevent its reinstallation. The historical

records, or "back-to-birth" traceability, for these LLPs must be maintained flawlessly, documenting every installation, removal, and accumulated cycle count throughout the component's entire service life. Discrepancies in the LLP records or the loss of traceability documentation will result in the immediate grounding of the component. Furthermore, if a brake system component is deferred under the Minimum Equipment List (MEL) or Configuration Deviation List (CDL), the specific operational restrictions, such as reduced landing weights or increased landing distance requirements, must be clearly placarded in the cockpit and documented in the aircraft's technical log, ensuring that the flight crew is fully aware of the degraded braking performance and the associated safety margins.

Appendix D: Daily Landing Gear & Brake Maintenance Checklist

This checklist governs trunnion alignments, wear tolerances, shock strut heights, and pressures required during wheel-end overhauls and daily landing gear inspection walks.

Table 12.1: Landing Gear Structural and Brake Specifications

Maintenance Item	Measurement / Range	Regulatory / Tool Ref
Main Gear Shock Strut Height	4.50 inches $\pm$ 0.25 in	Static load height measurement
Main Gear Strut Gas Charge	250 psi (1.7 MPa)	Dry nitrogen gas charge
Nose Gear Shock Strut Height	3.25 inches $\pm$ 0.25 in	Static load height measurement
Nose Gear Strut Gas Charge	180 psi (1.2 MPa)	Dry nitrogen gas charge
Cleveland 30-89 Disc Min Thickness	$\geq$ 0.235 inches (5.97 mm)	Dial Micrometer check
Cleveland 30-103 Disc Min Thickness	$\geq$ 0.312 inches (7.92 mm)	Dial Micrometer check
Brake Caliper Cylinder Bore Wear	$\leq$ 0.002 inches over nominal	Cylinder Bore Dial Indicator
Brake Wear Pin Indicator	Protrudes above caliper guide face	Checked with brakes pressurized
Anti-Skid Wheel Slip Ratio	10% to 15%	Optimal μ -slip controller bound
Up-lock Hook clearance	0.040 to 0.080 inches (1.0 to 2.0 mm)	Up-lock hook eccentrics check
Emergency Release Cable Slack	0.125 inches (3.17 mm)	Turnbuckle mechanical adjustment
Nose Steering Horn Left/Right Limit	30° angular deflection	Right/Left steering rod clevis
Centering Cam Spring Tension	50 lbs-in preload	Steering centering collar cart
Thermal Fuse Plug Melting Point	300°C (572°F)	Automatic wheel depressurization
Landing Gear Retract Swing Time	$\leq$ 10 seconds	Normal swing cycle on jacks
Brake Tie-Bolt Torque (Cleveland)	250 in-lbs (28.2 Nm) to 600 in-lbs	Threaded wheel halves assembly